The Compensation of Conscience

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Abstract

Markets are often lauded for their efficiency, but Nora Szech's research shows that individuals are more likely to act against their moral convictions in market contexts. Markets may insulate one from the moral demands of Adam Smith's *impartial spectator*, severing the link between personal responsibility and the moral outcomes of one's actions. Drawing on the literature of compensating wage differentials, we aim to quantify the *price of conscience* in labor market transactions. Using data from the National Longitudinal Survey of Youth 1997 (NLSY97) and Occupational Information Network (O*NET), we estimate the wage effects of occupations that varyingly require workers to act *against their sense of right and wrong.* Our longitudinal fixed-effects analysis shows that jobs demanding greater compromise of one's conscience offer higher wages, indicating a compensating differential for the burdening of conscience. This premium appears to increase with education and age, suggesting that experienced and educated workers require a higher price for the compromise of conscience. These findings extend Szech's experimental evidence on morals and markets to real-world labor markets, showing that the moral costs of market participation find their way into prices. As workers are paid more for compromising their convictions, markets may incentivize individuals to accept roles they otherwise find objectionable, aligning with Szech's research on how markets erode moral convictions and with Adam Smith's foundational observations on the compensation of dishonor and disgrace in labor markets.

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1 Introduction

Markets are often praised for their capacity to harness amoral self-interest by virtue of their capacity "to generate information about scarcity and allocate resources efficiently" (Falk and Szech, 2013). The late Nora Szech's research, particularly in her collaboration with Armin Falk in "Morals and Markets" (Falk and Szech, 2013), shows that markets are not morally neutral, instead inducing individuals to act against their own moral standards. They find that individuals are significantly more likely to make morally questionable decisions – such as allowing harm to a defenseless animal – when such decisions occur in a market context. The market properties of competition and bargaining appear to insulate market participants from acting on their moral convictions, enabling the abrogation of personal responsibility for the harms their actions may cause others. Szech's research challenges the philosophical view that markets are morally indifferent, highlighting instead how markets absorb, shape, and possibly degrade the moral sensibilities of market participants. Our research extends these experimental insights into real-world labor markets. Specifically, we examine how markets may differentially compensate jobs that require workers to compromise their moral *sense of right and wrong*, allowing for the quantification of the price of conscience in labor market transactions.

Workers face employment opportunities that may require some level of moral compromise, perhaps involving a task that imposes measurable harm on others or that violates a worker's own moral commitments. Akin to disamenities in the workplace like exposure to a health hazard, the requirement to compromise one's conscience may carry a compensating wage differential. In *The Wealth of Nations*, Smith (1776) recognized that workers are compensated for the loss of honor or the burden of disgrace. He writes: "Honour makes a great part of the reward of all honourable professions. In point of pecuniary gain, all things considered, they are generally under-recompensed... Disgrace has the contrary effect. The trade of a butcher is a brutal and an odious business; but it is in most places more profitable than the greater part of common trades. The most detestable of all employments, that of public executioner, is, in proportion to the quantity of work done, better paid than any common trade whatever." In the *Poverty of Philosophy*, Marx (1847) too noted that "virtue, love, conviction, knowledge, conscience" are lamentably tradable objects and priced in markets.

Szech's extensive experimental and theoretical work on moral decision-making in markets, including her studies on moral self-licensing (Engel and Szech, 2020), strategic moral ignorance (Serra-Garcia and Szech, 2022), and the diffusion of moral responsibility in group settings (Falk et al., 2020) shows that moral considerations pervade market behavior. Falk and Szech (2013) note: "Many people express objections against child labor, other forms of exploitation of the workforce, detrimental conditions for animals in meat production, or

environmental damage. At the same time, they seem to ignore their moral standards when acting as market participants ... thereby consciously or subconsciously creating the undesired negative consequences to which they generally object."

With data from the National Longitudinal Survey of Youth 1997 (NLSY97) on worker wages coupled with data from the Occupational Information Network (O*NET) on the extent to which workers in various occupations are routinely asked "to go against their sense of right or wrong", we analyze the pricing of conscience in labor markets or how, for a price, workers may "ignore their moral standards when acting as market participants"(Falk and Szech, 2013). Together, NLSY97 and O*NET data provide a rich panel (1997-2021) of job-taking behaviors and wage outcomes to undertake our analysis. Across a series of tests, we show that jobs with lower conscience scores – those requiring greater compromise of conscience – tend to offer higher wages, consistent with the theory of equalizing differences in Brown (1980), but in our case involving workers receiving measurable wage premiums for the acceptance of jobs with higher moral burden. We also find that the price of conscience is heterogeneous, appearing to increase with both education attainment and age. Absent the compensation of conscience, worker willingness to endure the moral costs of markets noted by Szech appear to decrease with age. Like other disamenities, we find evidence of a nonlinear relationship between worker compensation and occupational conscience scores, suggesting that the premium for the relative presence of compromise is greater than the penalty (or non-wage benefit) for the relative absence of compromise.

These findings support the notion that labor markets place a price on the compromise of conscience, suggesting that workers are indeed compensated for the moral costs they bear as market participants. Whereas Szech's controlled experiments provide causal evidence that markets erode moral convictions, our study shows that these convictions find their way into prices, possibly inducing workers to assume occupations that they might otherwise find objectionable.

Economists have focused their attention on the price labor markets place on workplace (dis)amenities, such as health insurance, maternity benefits, sexual harassment, schedule flexibility, job stress and duration, and the risk of harm (e.g., Olson, 1981; Arnould and Nichols, 1983; Dickens, 1984; Garen, 1988; Moore and Viscusi, 1989; Ruser, 1991; Fairris, 1992; Martinello and Meng, 1992; Gronberg and Reed, 1994; Gruber, 1994; French and Dunlap, 1998; Meng and Smith, 1999; Gunderson and Hyatt, 2001; Lanfranchi et al., 2002; Leeth and Ruser, 2003; Miller, 2004; Villanueva, 2007; Dey and Flinn, 2005; McCrate, 2005; Lehrer and Pereira, 2007; Hersch, 2011; Qin and Chernew, 2014; Mas and Pallais, 2017; Guardado and Ziebarth, 2019; Evans and Taylor, 2020; Hoffmann et al., 2020; Lavetti, 2020;

Anderson, 2022; Charles et al., 2022; Sparber and Zavodny, 2022).[1] Of particular relevance to our research question on the price of conscience are studies on the compensation of *other-regarding occupations*. Such studies find that workers involved in the provision of care are paid less. Such workers are said to suffer a measurable *care wage penalty*, with the down pressure on wages offset by the intrinsic rewards realized in the care of others (England et al., 2002; Hirsch and Manzella, 2015; Budig et al., 2019). Similarly, a number of other studies investigate whether nonprofit workers earn less than for-profit workers (Handy and Katz, 1998; Leete, 2001; Ruhm and Borkoski, 2003; Mocan and Tekin, 2003; Jones, 2015).[2] We bridge this literature on compensating differentials in the labor market with the *economics of morality* highlighted in Szech's research. While the welfare implications of our results are difficult to ascertain, they provide a glimpse of how markets systematically reward workers for doing things they define as wrong.

The rest of our paper proceeds as follows: Section 2 describes our data and empirical methods, including how we construct the conscience intensity of occupations using O*NET data to explain wages in the NLSY97. Section 3 presents the results of our longitudinal fixed-effects analysis. Finally, Section 4 discusses the implications of these results, concluding with our final remarks.

2 Data and Methods

2.1 Sample Data

Our analysis draws on two data sources. First, we use publicly available data from the 1997-2021 waves of the NLSY97, administered by the Bureau of Labor Statistics (BLS). This survey is a nationally representative sample of 8,984 individuals born between 1980 and 1984, who were 13 to 17 years old at the time of the first interviews in 1997, and 37 to 41 years old in the most recent round in 2021. Respondents in the NLSY97 were interviewed annually from 1997 to 2011 and biennially thereafter – i.e., in 2013, 2015, 2017, 2019, and 2021. The NLSY97 provides detailed information on respondents, including demographic characteristics, family background, education, employment, income, health, and family formation. The survey tracks job history, wages, and occupational details. A key advantage of the NLSY97 over other datasets for our analysis, such as the Current Population Survey

[1] See Lavetti (2023) for a comprehensive review of the compensating wage differentials literature.

[2] These studies yield mixed evidence. For example, using data on individuals aged 25 to 55 from the 1988–1994 CPS-ORG, Jones (2015) find that employees in nonprofit organizations receive virtually the same wages as comparable employees in profit-seeking companies. In contrast, Ruhm and Borkoski (2003) shows that a longitudinal analysis focusing on movements of workers between nonprofit and profit-seeking employers suggests a nonprofit penalty of between 2 and 4 percent.

Outgoing Rotation Group (CPS-ORG), is it allows us to track all employment instances for each respondent using unique job identifiers. This feature allows for multiple employment observations within years, offering detailed insights into respondents' labor supply and employment history over the life course. A disadvantage of the sample is truncation with respect to age. As noted previously, since all respondents in the sample were born during the years 1980 through 1984, the eldest of respondents were 41 years of age in 2021.

The second data source we use is O*NET, administered by the Employment & Training Administration (ETA) within the U.S. Department of Labor (DOL). O*NET is a comprehensive database that provides extensive ratings on the skills and tasks required for different occupations, as well as the work environment characteristics associated with each occupation. O*NET provides insights into general work activities and the context in which tasks are performed, including physical and social factors that define the work environment. Personal attributes and work values – such as dependability, attention to detail, and achievement orientation – are also profiled, offering a well-rounded view of the traits conducive to success in various occupational roles. O*NET was developed to replace the Dictionary of Occupational Titles (DOT), which was challenging to update and faced criticism concerning its sampling strategy, content coverage, and measurement methods. Compared to the DOT, O*NET incorporates improved sample sizes, sampling techniques, and content coverage (Handel, 2016).

After carefully merging these datasets, our final sample includes 481 unique occupations and 107,312 worker-occupation-year observations, consisting of workers aged 18-41 who work full-time (defined as 30 or more hours per week). The average number of occupation-year observations per worker is approximately 19 while the average number of unique occupations held per worker is 9. In the following section, we provide more detailed information on the variables used in our analysis.

2.1.1 Outcome Variable

Our outcome variable in all analyses is the logarithm of real hourly compensation in the NLSY97. This variable includes not only base hourly wages but also additional earnings such as overtime pay, tips, and bonuses. This is likely to be important since there may be differences in pay across occupations that are not captured exclusively by wage rates. Throughout the paper, we will use the terms compensation and wages interchangeably. We converted reported hourly pay rates into 2020 U.S. dollars.[3] Since some observations include unrealistically low or high compensation values, we followed Dey et al. (2021) and truncated

[3]Yearly adjustment factors were obtained via the *cpigen* command in Stata.

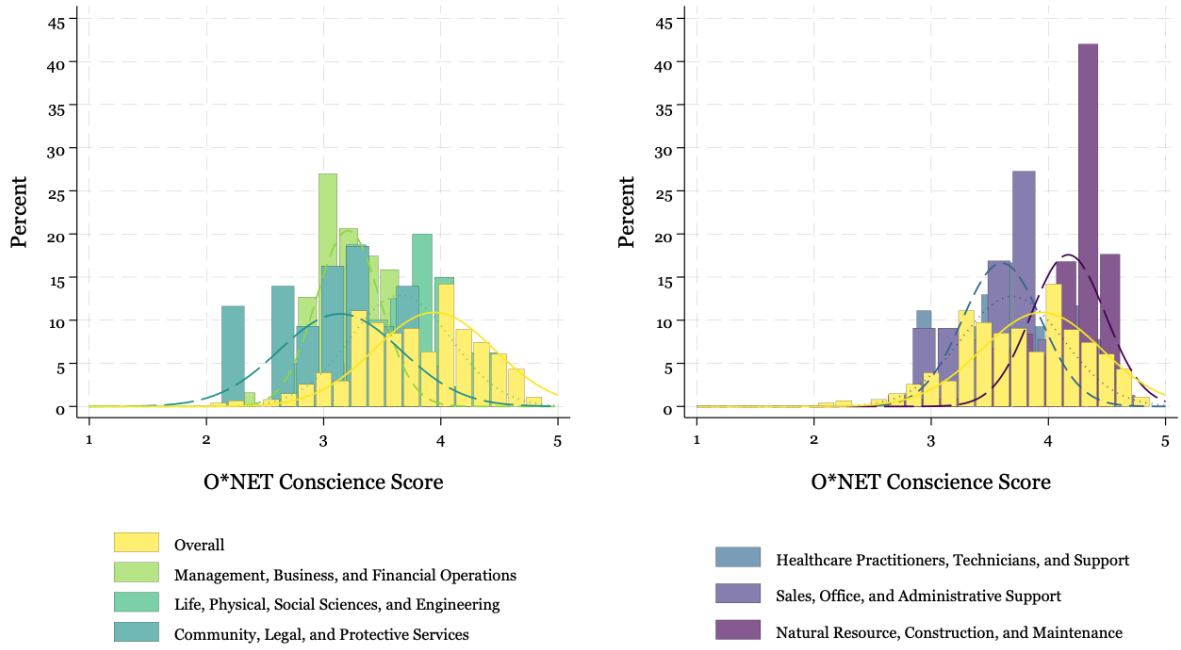
the compensation distribution, limiting it to hourly compensation between the real federal minimum wage ($ USD 2020) and an upper bound of $150.00.

2.1.2 Explanatory Variable

Our explanatory variable of interest is the *conscience score* of occupations. This variable is from the O*NET's work values component within worker characteristics, where incumbents of occupations are asked: *"Workers on this job are never pressured to do things that go against their sense of right and wrong."* Responses are scored from 1 to 5, with 5 meaning a typical worker is never asked to compromise their conscience in a job. This measure captures the degree to which workers in various occupations can act according to their internal moral standards without pressure to compromise their *sense of right and wrong*. Taking from Smith (1759), we refer to this measure as "conscience." In *The Theory of Moral Sentiments*, Smith (1759) describes conscience as the "great judge and arbiter of our conduct," an internal mechanism that allows one to evaluate the propriety or impropriety of their actions. Smith famously argued that conscience functions through the perspective of the impartial spectator, guiding individuals to align their behavior with their sense of right and wrong.

The main challenge in merging the O*NET conscience scores with the NLSY97 sample is that O*NET uses the 2018 Standard Occupational Classification (SOC) system, while the NLSY97 uses occupational codes based on the 2002 Census classification. We used the crosswalk provided by the Integrated Public Use Microdata Series (IPUMS) to link the O*NET conscience scores to the NLSY97 sample (see Ruggles et al., 2020). Once the crosswalk is used to match 2002 Census occupations to 2018 SOC O*NET occupations, we then collapsed on the 2002 Census occupations using the mean assignment.

While most occupation matches were straightforward, some required manual inspection. For example, the 2002 Census occupation "Emergency Medical Technicians and Paramedics" (3400) should correspond to the 2018 SOC occupations "Emergency Medical Technicians" (292042) and "Paramedics" (292043). However, these 2018 SOC occupations are absent in the O*NET data. Instead, "Emergency Medical Technicians and Paramedics" (3400) is matched to the 2018 SOC occupation "Firefighters" (332011). Similar discrepancies exist for other NLSY97 occupations; a complete list of these manually matched occupations is provided in Table A1 in the appendix. Moreover, some NLSY97 occupations correspond to multiple O*NET occupations and vice versa. A pertinent example is the 2002 Census occupation "Physicians and Surgeons". In the O*NET data, Neurologists, Obstetricians, Pediatricians, Hospitalists, and Urologists are defined as separate occupations. In such cases, each detailed 2018 SOC O*NET occupation is assigned to the single "Physicians and Surgeons" 2002 Census occupation.



Notes: The raw O*NET conscience score is used, where a score of 5 indicates that workers are never asked to compromise their conscience in their job. Sectors are defined according to the Census 2002 Occupation Codes.

Figure 1. Distribution of NLSY97 Occupations by Sector Across O*NET Conscience Score

Figure 1 shows the distribution of 481 unique occupations in the NLSY97 sample across the domain of the raw O*NET conscience score by sector. We divided Figure 1 into two panels for easier visualization, each showing the distribution for three occupational groups alongside the overall distribution of the O*NET conscience score. The overall distribution of conscience scores across all 481 unique occupations is slightly skewed to the right, with most occupations clustering around scores of 3 and 4 (with a mean of 3.79 and a standard deviation of 0.53). The left panel of Figure 1 displays the distribution of conscience scores for occupations in 1) Management, Business, and Financial Operations, 2) Life, Physical, Social Sciences, and Engineering, and 3) Community, Legal, and Protective Services. The Management, Business, and Financial Operations group distribution peaks around a score of 3 (with a mean of 3.21 and a standard deviation of 0.27). Life, Physical, Social Sciences, and Engineering occupations peak around scores of 3 and 4 (with a mean of 3.66 and a standard deviation of 0.43). Community, Legal, and Protective Services occupations show a broader spread with concentrations around a score of 3 (with a mean of 3.14 and a standard deviation of 0.52).

The right panel of Figure 1 presents the distribution for 1) Healthcare Practitioners, Technicians, and Support, 2) Sales, Office, and Administrative Support, and 3) Natural Resource, Construction, and Maintenance occupations. Healthcare Practitioners, Technicians, and

Support show clustering around scores of 3 and 4 (with a mean of 3.60 and a standard deviation of 0.34). Similarly, Sales, Office, and Administrative Support occupations show a peak around scores of 3 and 4 (with a mean of 3.69 and a standard deviation of 0.44). Resource, Construction, and Maintenance occupations, by contrast, have a higher concentration of scores at 4 and above (with a mean of 4.17 and a standard deviation of 0.32).

To form a sense of the types of occupations within each sector by O*NET conscience score, we present the top and bottom 10 occupations ranked by their scores in Table 1. Consider the list of Community, Legal, and Protective Service occupations. Lawyers, judges, magistrates, and arbitrators must oblige professional ethics and guidelines that routinely conflict with their own sense of right and wrong. A common ethical dilemma of a criminal defense attorney, for example, involves representation of a guilty client. In such cases, the attorney must set aside their personal misgivings, assuring that their client's rights are protected. By contrast, municipal firefighters, lifeguards, and court reporters report significantly less demands on their conscience in the performance of their jobs.

Table 1. Top and Bottom 10 Occupations by Sector and Ranked by O*NET Conscience Score

Top Occupations			Bottom Occupations		
Rank	Title	Score	Rank	Title	Score
	Management, Business, and Financial Operations (0010-0950)				
1	Agents of Artists, Performers, and Athletes	2.25	53	Computer and Information Systems Managers	3.50
2	Environmental Compliance Inspectors	2.75	53	Training and Development Managers	3.50
3	Tax Collectors and Revenue Agents	2.87	53	Job Analysis Specialists	3.50
3	Advertising and Promotions Managers	2.87	53	Architectural and Engineering Managers	3.50
3	College Administrators	2.87	53	Food Service Managers	3.50
3	Loan Officers	2.87	53	Event Planners	3.50
3	Financial Analysts	2.87	53	Treasurers and Controllers	3.50
3	Gaming Managers	2.87	60	Wholesale and Retail Buyers	3.62
9	Chief Executives	2.88	61	Postmasters	3.75
10	Auditors	3.00	62	Nursery and Greenhouse Managers	4.00
	Life, Physical, Social Sciences, and Engineering (1300-1960)				
1	Political Scientists	2.62	71	Electro-Mechanical Technicians	4.25
1	Clinical Psychologists	2.62	71	Surveying Technicians	4.25
3	Urban and Regional Planners	2.75	71	Hydrologists	4.25
3	Forensic Science Technicians	2.75	71	Civil Drafters	4.25
5	Counseling Psychologists	2.87	71	Architectural Drafters	4.25
6	School Psychologists	3.00	76	Atmospheric and Space Scientists	4.37
6	Industrial Safety and Health Engineers	3.00	76	Mapping Technicians	4.37
8	Foresters	3.12	76	Geoscientists	4.37
8	Industrial Psychologists	3.12	76	Mechanical Drafters	4.37
10	Epidemiologists	3.25	80	Astronomers	4.50
	Community, Legal, and Protective Services (2000-2150, 3700-3950)				

Continued on next page

Top Occupations			Bottom Occupations		
Rank	Title	Score	Rank	Title	Score
1	Arbitrators, Mediators, and Conciliators	2.12	34	Fire Inspectors	3.50
1	Administrative Law Judges	2.12	34	Animal Control Workers	3.50
1	Lawyers	2.12	34	Title Examiners and Abstractors	3.50
4	Judges and Magistrates	2.25	37	Police Records Officers	3.62
4	Probation Officers	2.25	38	Court Reporters	3.75
6	Child, Family, and School Social Workers	2.62	38	Lifeguards, Ski Patrol	3.75
7	Mental Health Counselors	2.75	38	Security Guards	3.75
7	Private Detectives and Investigators	2.75	38	Forest Fire Inspectors	3.75
9	Clergy	2.87	42	Municipal Firefighters	4.00
9	Police Detectives	2.87	43	Crossing Guards	4.25
	Healthcare Practitioners, Technicians, and Support (3000-3650)				
1	Surgeons	2.87	45	Physical Therapist Assistants	3.87
2	Family and General Practitioners	3.00	46	Cardiovascular Technicians	4.00
2	Internists, General	3.00	46	Radiation Therapists	4.00
2	Psychiatrists	3.00	46	Medical Assistants	4.00
2	Obstetricians and Gynecologists	3.00	46	Orthotists and Prosthetists	4.00
2	Pediatricians, General	3.00	50	Medical Equipment Preparers	4.12
7	Athletic Trainers	3.12	50	Dental Hygienists	4.12
8	Psychiatric Technicians	3.25	50	Radiologic Technicians	4.12
8	Registered Nurses	3.25	50	Surgical Technologists	4.12
8	Occupational Health and Safety Specialists	3.25	54	Medical Records Technicians	4.25
	Sales, Office, and Administrative Support (4700-5930)				
1	Sales Representatives, Wholesale	2.62	68	Marking Clerks	4.25
2	Bill and Account Collectors	2.87	68	Mail Clerks, Except Postal Service	4.25
2	Door-To-Door Sales and Street Vendors	2.87	68	File Clerks	4.25
2	Telemarketers	2.87	71	Proofreaders and Copy Markers	4.37
5	Retail Salespersons	3.00	71	Order Fillers, Wholesale and Retail Sales	4.37
5	Payroll and Timekeeping Clerks	3.00	71	Office Machine Operators	4.37
5	Sales Agents, Securities	3.00	71	Stock Clerks	4.37
8	Parts Salespersons	3.12	71	Insurance Claims Clerks	4.37
8	Bookkeeping and Auditing Clerks	3.12	71	Shipping and Receiving Clerks	4.37
8	Demonstrators and Product Promoters	3.12	77	Data Entry Keyers	4.75
	Natural Resource, Construction, and Maintenance (6000-7620)				
1	Construction and Building Inspectors	2.75	110	Pipe Fitters and Steamfitters	4.50
2	Hunters and Trappers	3.00	110	Fence Erectors	4.50
3	Agricultural Inspectors	3.25	110	Electrical Power-Line Installers and Repairers	4.50
4	Supervisors of Animal Care Workers	3.50	110	Floor Sanders and Finishers	4.50
5	Fishers and Related Fishing Workers	3.62	110	Tire Repairers and Changers	4.50
5	Supervisors of Trades and Extraction Workers	3.62	110	Refrigeration Mechanics and Installers	4.50
5	Supervisors of Logging Workers	3.62	116	Musical Instrument Repairers and Tuners	4.59
5	Animal Breeders	3.62	117	Watch Repairers	4.62
9	Construction Laborers	3.75	117	Pile-Driver Operators	4.62
9	Hazardous Materials Removal Workers	3.75	117	Logging Equipment Operators	4.62

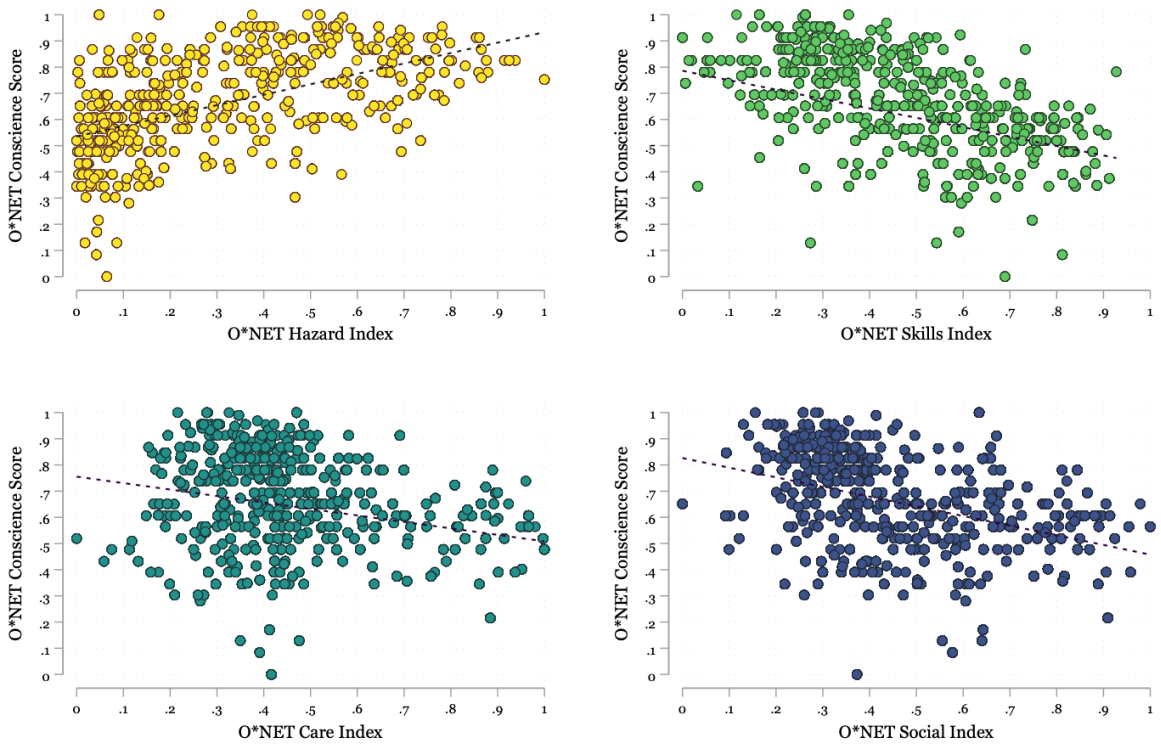
Note: Some occupational titles are shortened without loss of meaning. Sector code ranges are in parentheses and correspond to Census 2002 Occupation Codes.

2.1.3 Control Variables

We also include controls from the NLSY97 sample to account for human capital, labor supply, and occupational characteristics that are known to jointly determine hourly compensation. These controls include indicators for enrollment status and highest degree obtained, as well as measures for cumulative years of unemployment, cumulative years not-in-the-labor-force, usual weekly hours worked, and years of job-specific tenure. We also account for job characteristics with indicators for union membership and self-employment status.

To capture occupation-specific factors that may also influence wages, we construct five additional control variables from the O*NET database. We apply Principal Component Analysis (CPA) to create indices for four variables: a hazard index (incorporating indicators of bodily harm risk, such as exposure to contaminants, hazardous conditions, hazardous equipment, high places, minor burns, cuts, bites or stings, and whole-body vibration), a skills index (capturing competencies required across occupations, including reading comprehension, active listening, writing, speaking, mathematics, and science), a social index (reflecting tasks performed for others or involving personal connections on the job, such as social service and social orientation), and a care index (capturing the importance of assisting and caring for others, as well as concern for others). Finally, we include a measure of occupational autonomy intensity that captures the level of supervision a worker experiences, using a working definition that asks incumbents of occupations: "*How much decision making freedom, without supervision, does the job offer?*". Responses are scored from 1 to 5, with 5 meaning a typical worker has a lot of freedom in a job. We scale these occupation variables to range from zero to one using min-max standardization to aid in the interpretation of our empirical estimates.[4]

Figure 2 shows the correlations between the conscience score and various O*NET indices. There is a moderately strong positive correlation between the conscience score and the hazard index, suggesting that more hazardous jobs tend to allow workers to stay aligned with their conscience more frequently. In contrast, the skills index shows a moderately strong negative correlation with the conscience score, indicating that higher-skill jobs are somewhat associated with greater moral compromise. Moreover, the care and social indices also exhibit negative correlations with the conscience score, implying that jobs emphasizing care for others and social interaction may require slightly more moral compromise. Overall, the observed trends hint at how different job attributes might relate to the extent of moral compromise workers perceive in their roles.



Notes: Indices and scores are from the O*NET database and scaled from zero to one using min-max standardization. A value of 0 indicates the lowest level of the respective attribute.

Figure 2. Correlations between O*NET Conscience Score and other O*NET Indices

2.1.4 Sample Descriptive Statistics

Table 2 presents our sample descriptive statistics for the main variables in the NLSY97 sample. The average real hourly compensation is approximately \$19.50 per hour, with values ranging from around \$7 to \$150 per hour. Moreover, about 13% of respondents reported being enrolled in K-12, college, or a graduate program, while the majority (74%) reported having a high school education or less. The average cumulative time spent unemployed is roughly 0.74 years (or about 9 months), with a standard deviation of 1.03 years. This suggests that while most respondents experience relatively short periods of unemployment, there is significant variability, with some facing prolonged unemployment. On average, respondents have spent about 2.73 years (approximately 33 months) not in the labor force, with a standard deviation of 2.26 years. Approximately 9% of respondents are union members, and about 4% are self-employed. The average workweek is 40.47 hours, with the 25th and 75th percentiles both at 40 hours, indicating that a standard 40-hour workweek is common in the sample. Job tenure averages 2.51 years, ranging from less than a year to nearly 30 years, reflecting diverse levels of job stability and experience among respondents.

[4]Formula: $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$, where x is an original value and x' is the standardized value.

Table 2. Sample Descriptive Statistics

	Mean	St. Dev.	Min	25th	75th	Max	N
Outcome and Explanatory Variables							
Hourly Compensation ($ USD 2020)	19.47	13.41	6.93	11.61	22.17	149.99	107,312
Conscience Score	0.63	0.17	0.00	0.52	0.77	1.00	107,312
Human Capital and Job Characteristics							
Enrollment Status	0.13	0.34	0.00	0.00	0.00	1.00	107,312
Degree							
$\leq$ High School	0.74	0.44	0.00	0.00	1.00	1.00	107,312
Associates	0.06	0.24	0.00	0.00	0.00	1.00	107,312
Bachelors	0.16	0.37	0.00	0.00	0.00	1.00	107,312
Graduates	0.05	0.21	0.00	0.00	0.00	1.00	107,312
Cum. Years NILF	2.73	2.26	0.00	1.10	3.77	22.52	107,312
Cum. Years in Unemployment	0.74	1.03	0.00	0.10	0.98	17.79	107,312
Union Status	0.09	0.29	0.00	0.00	0.00	1.00	107,312
Self-Employed	0.04	0.19	0.00	0.00	0.00	1.00	107,312
Work Hours per Week	40.47	7.77	30.00	40.00	40.00	168.00	107,312
Job Tenure (Years)	2.51	3.16	0.00	0.44	3.27	29.92	107,312
Other Occupational Attributes							
Hazard Index	0.25	0.21	0.00	0.08	0.39	1.00	107,312
Skills Index	0.43	0.22	0.00	0.26	0.60	1.00	107,312
Care Index	0.49	0.18	0.00	0.37	0.56	1.00	107,312
Social Index	0.52	0.19	0.00	0.35	0.66	1.00	107,312

Notes: O*NET indices and score are scaled to range from zero to one using min-max standardization. NILF implies not-in-the-labor-force. The sample is restricted to workers aged 18 and older who work at least 30 hours per week, with hourly wages between the real federal minimum wage (in 2020 USD) and an upper bound of $150.00. Source: NLSY97, Waves 1997–2021.

Although not reported in Table 2, about 54% of respondents are men, 25% identify as Non-Hispanic Black, and 21% identify as Hispanic. Furthermore, 79% of respondents reside in urban areas, and about 40% live in the South of the U.S. in our final sample. In the following section, we outline our estimation strategy.

2.2 Estimation Strategy

Empirically quantifying compensating differentials for various workplace (dis)amenities presents several challenges in real-world labor markets, primarily due to potential confounding factors that correlate with both preferences for (dis)amenities and wages (Lavetti, 2023). Our estimation strategy is guided by the theoretical and empirical frameworks for studying compensating differentials developed by Rosen (1974, 1986) and Brown (1980).

Rosen (1974, 1986) built upon Smith (1776)'s proto-theory of compensating differentials by formalizing the concept within a hedonic wage framework, positing that wages adjust to reflect the relative desirability of job attributes. In Rosen (1986)'s model, workers and employers engage in a market where wages reflect both the skill level and the amenity preferences of the workforce. In competitive labor markets, job characteristics and wages coalesce to a matching equilibrium where workers select occupations that maximize their utility, bal-

ancing both wage and non-wage job characteristics. However, Brown (1980) emphasized the importance of accounting for omitted worker characteristics in estimating the compensating differentials associated with undesirable job characteristics. Specifically, Brown (1980) highlighted that unobserved factors related to worker ability and personal traits can bias the estimates of compensating differentials. To address this, longitudinal data are essential to properly control for unobserved heterogeneity among workers.

According to the theory of equalizing differences described by Brown (1980)[5], workers pursue job opportunities based on their personal characteristics (X), wages on offer (w), and non-wage job attributes (Z). In this framework, employers must offer higher wages for jobs with less desirable attributes to draw workers with given characteristics. Workers maximize utility by combination of wages and job conditions. In this context, Z is a vector of non-wage job characteristics, with larger values indicating less desirable job attributes. The function $f(w, Z; X)$ describes the relationship between wages, job attributes, and worker characteristics. The theory implies that the marginal compensation for undesirable job attributes must be positive: $\frac{\partial w}{\partial Z_j} > 0$ for all j, indicating that jobs with more negative attributes must offer higher wages to compensate workers for the job burdens assumed.

2.2.1 Cross-Sectional Model

To translate this theoretical insight into our empirical framework, we specify a semi-logarithmic wage equation, which is common in the literature on compensating wage differentials, as follows:

$$ln(w_{ijt}) = X'_{ijt}\beta + \alpha Z_{ij} + \delta_t + \epsilon_{ijt}, \tag{1}$$

where $ln(w_{ijt})$ is the natural logarithm of hourly compensation for worker i in occupation j at time t, X'_{ijt} represents a vector of observed worker-specific characteristics (see Table 2), and Z_{ij} captures the job attributes. In our context, Z_{ij} is the observed O*NET conscience score.[6] The coefficient of interest, α, quantifies the compensating differentials for conscience compromised occupations. We include year fixed-effects, δ_t for a pooled cross-sectional estimate of conscience score on wages. ϵ_{ijt} is an error term assumed to be iid or come from a low-order moving-average process.

[5] In this literature, researchers have defined the concept of "compensating wage differentials" in various ways. We follow Lavetti (2023) in defining the terms "compensating wage differentials" and "equalizing wage differentials" synonymously.

[6] The O*NET values assigned to each occupation are fixed over time. O*NET updates ratings for occupations on a rolling basis; it takes several years for all occupations to have revised ratings. Therefore, Z_{ij} does not have a t subscript in our context.

In cross-sectional models such as Equation (1), if certain individual characteristics such as ability or motivation, are not included in X'_{ijt}, the estimated coefficient α could be biased. Workers with higher unobserved abilities may be more likely to sort into occupations with high or low occupational conscience scores, leading to biased estimates of compensating differentials. This reflects Brown (1980)'s concern that the omission of individual-specific attributes in cross-sectional studies can result in incorrect inferences about compensating wage differentials.

2.2.2 Worker Fixed-Effects Model

To mitigate the bias introduced by unobserved worker characteristics, we extend model (1) by incorporating worker fixed-effects. This approach allows us to control for time-invariant worker-specific factors, which helps reduce the omitted variable bias discussed in the cross-sectional model. The model is specified as follows:

$$ln(w_{ijt}) = X'_{ijt}\beta + \alpha Z_{ij} + A'_{ij}\gamma + \eta_i + \delta_t + \epsilon_{ijt}, \tag{2}$$

where η_i represents worker fixed-effects, capturing all time-invariant characteristics specific to each worker, such as innate ability, motivation, and other personal traits that may influence compensation. By including η_i, we ensure that our estimates of α are less likely to be confounded by unobserved heterogeneity, addressing the concerns raised by Brown (1980). The year fixed-effect δ_t controls for time-varying shocks that are common across all workers – such as wage shocks resulting from large-scale economic downturns, and A'_{ij} accounts for a vector of other O*NET occupational attributes that are known to influence wage outcomes.[7]

Because the O*NET conscience scores assigned to each occupation remain fixed over time, the identification of α and β coefficients in the fixed-effects model relies on within-worker variation from switching in and out of occupations over time. Thus, changes in Z_{ij} and X'_{ijt} for a given worker captures the compensating differential. By identifying worker-specific wage changes resulting from movements across years into or out of occupations with different levels of conscience score, we can account for bias due to worker heterogeneity correlated with conscience scores.

Even with worker fixed-effects, certain challenges remain. As Brown (1980) noted, while longitudinal data improve our ability to control for unobserved, time-invariant worker char-

[7] We also include indicators for geography and industry. For geography, we control for urbanicity (urban, rural, or unknown), metropolitan statistical area (MSA) status (not in MSA, in MSA but not in central city, in MSA and in central city, or in MSA with central city location unknown), and broad Census region (Northeast, North Central, South, and West). Moreover, we include a set of indicators for industry to capture any industry-specific factors that might affect wages, such as variations in labor market conditions, demand for specific skills, or industry-specific regulations.

acteristics, it does not fully eliminate the potential for bias due to time-varying residual confounders. In our context, the strength of unobserved confounding variables in driving the observed relationship between conscience scores and wages cannot be directly tested. However, time-varying controls related to human capital and job characteristics – such as enrollment status, highest degree obtained, cumulative years of unemployment, cumulative years not-in-the-labor-force, usual weekly hours worked, years of job-specific tenure, union membership, and self-employment status – are included in all analyses. This helps account for broader changes in labor-market decisions (i.e., selection into (out of) conscience compromised occupations) at the worker level that are correlated with both conscience scores and wages. Despite these efforts, residual confounding from unobserved time-varying factors may still persist. Consequently, our coefficient estimate α should be interpreted with caution as a genuine causal effect.

There are two limitations in using O*NET job skill/task measures. First, as previously mentioned, the O*NET values assigned to each occupation remain fixed over time. Second, the value of each O*NET attribute does not vary among workers within the same occupation. The first issue is less concerning because the relative skill and task attributes of occupations tend to change gradually. With respect to the second issue involving the measurement of job attributes at the occupation level rather than the worker level, it is unclear whether and to what extent measurement error in O*NET job attributes is addressed through worker fixed-effects. We conservatively cluster standard errors at the occupation level, as suggested by Brown (1980) and Abadie et al. (2023), to account for correlation in error terms due to common job characteristics within occupations.

Finally, a functional form mis-specification generally means that the model does not account for important nonlinearities, causing bias in the remaining parameter estimates. To address this challenge, we consider a piecewise linear worker fixed-effects model that allows for different slopes across various ranges of the conscience score in Section 3.

3 Results

This section presents our empirical results in segments. First, we discuss the baseline relationship between the conscience score and wages. Second, we present sensitivity analyses to assess the robustness of our findings. Third, we examine heterogeneous effects by age and education. Finally, we relax the assumption of a strictly linear relationship between conscience scores and wages, examining potential nonlinear effects of conscience scores on wages.

3.1 Baseline Results

Our baseline results are presented in Table 3. The first column shows the association between the conscience score and hourly compensation, conditional on year fixed-effects. In the second column, we add controls from Table 2, along with additional factors – such as indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, race/ethnicity, sex, and age.[8] In the third column, we include worker fixed-effects but exclude the controls. The final column presents our fully saturated model, including both the controls and worker fixed-effects.[9]

In column (1), where only year fixed-effects are controlled for, the coefficient on the conscience score is -0.455 (or -36.6%).[10] This suggests that moving from an occupation where workers are most pressured to compromise their conscience (Conscience Score = 0) to one where they are least pressured (Conscience Score = 1) is associated with a 36.6% decrease in real hourly compensation. When we add controls from Table 2 in column (2), the coefficient decreases in magnitude to -0.177 (or -16.2%), though it remains statistically significant at the 1% level. Controls capture important confounding factors that account for part of the compensation outcome.

Table 3. Estimates of O*NET Conscience Score vis-a-vis Log Real Hourly Compensation

	(1)	(2)	(3)	(4)
Conscience Score (Z_{ij})	-0.455***	-0.177***	-0.156***	-0.091***
	(0.126)	(0.058)	(0.059)	(0.034)
Constant	2.711***	2.270***	2.385***	2.249***
	(0.101)	(0.073)	(0.050)	(0.052)
Observations	107,312	107,312	107,312	107,312
R-squared	0.207	0.454	0.614	0.665
Controls	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
Worker FE	No	No	Yes	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET conscience score is scaled from zero to one using min-max standardization. Controls are from Table 2. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, race/ethnicity, sex, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). Note that indicators for race/ethnicity and sex are omitted when controlling for worker fixed effects.

When we include worker fixed-effects in column (3), the coefficient on the conscience score further decreases in magnitude to -0.156 (or -14.4%), and the R-squared increases to about

[8] Age is grouped into 3-year intervals.

[9] For full model results, please refer to Table A2 in the appendix.

[10] Percentage change in real hourly compensation $\approx [exp(\hat{\alpha}) - 1] \times 100$, where $\hat{\alpha}$ is the estimated coefficient.

61.4%. In our preferred specification, presented in column (4), we include both the time-varying worker controls and worker fixed-effects. The coefficient on the conscience score further decreases in magnitude to -0.091 (or about -9%). Note that the R-squared value increases modestly from 61.4% in column (3) to 66.5% in column (4). This modest increase suggests that adding controls in column (4) only slightly enhances explanatory power beyond what is captured by worker fixed-effects. The relatively small change in R-squared increases our confidence that we are effectively controlling for unobservable, time-invariant factors, thereby reducing the potential for omitted variable bias.

Consistent with Brown (1980), our results in columns (2) and (4) underscore the importance of controlling for worker fixed-effects in a panel data setting. Brown (1980) shows that the compensating wage differential for jobs with bad working conditions is statistically significant in a cross-sectional model.[11] However, when worker fixed-effects are included, the coefficient reported by Brown (1980) shrinks by about 53% and becomes statistically insignificant. We observe a similar pattern in our results: with the inclusion of worker fixed-effects, our coefficient decreases by approximately 48% (from -0.177 in column (2) to -0.091 in column (4)). However, unlike Brown (1980), our coefficient remains statistically significant. Controlling for time-invariant factors is crucial in our context because workers who prioritize their conscience in occupational decisions may possess unobserved traits correlated with lower wages – such as a preference for job satisfaction. If these unobserved traits are not controlled for, they could lead to an overestimation of the compensation effects of the conscience score.

Our results confirm a consistent relationship between the conscience score and wages. Specifically, by exploiting within-worker variation in the conscience score and controlling for time-varying worker and occupational characteristics, and industry-specific factors, we find that workers in occupations where they are most often pressured to compromise their conscience witness an increase in wages.

3.2 Sensitivity Results

In this section, we check the sensitivity of the previously discussed results in Table 3 across an ensemble of tests. The first column restates our preferred specification from column (4) of Table 3 for ease of comparison, while the subsequent columns present various robustness checks.

First, we add only the autonomy score as an additional control variable. Autonomy at work is a critical aspect of the workplace environment, referring to the degree of control a

[11] Brown (1980) defines jobs with poor working conditions as those involving extremes of cold or heat, humidity, vibrations, or hazards, similar to our construction of the O*NET hazard index.

worker has over their working conditions and the nature of their tasks (DeVaro, 2006). It is plausible that workers with a high level of autonomy experience less pressure to compromise their conscience in a job. By controlling for workplace autonomy, we aim to isolate the independent effect of conscience score on hourly compensation. This control makes sure that the observed relationship between conscience score and hourly compensation is not confounded by variation in job autonomy. Specifically, the estimated conscience coefficient from column (2) in Table 4 remains consistent with the baseline model in column (1). Note that the R-squared value is also similar between columns (1) and (2). It is also worth noting that the number of observations decreases to 100,689 when controlling for autonomy, because autonomy scores are not available for all occupations in our final sample.

Table 4. Sensitivity Estimates of O*NET Conscience Score vis-a-vis Log Real Hourly Compensation

	(1)	(2)	(3)	(4)	(5)
Conscience Score (Z_{ij})	-0.091***	-0.089**	-0.092***	-0.077**	-0.084***
	(0.034)	(0.038)	(0.029)	(0.035)	(0.031)
Constant	2.249***	2.235***	2.720***	2.374***	2.760***
	(0.052)	(0.057)	(0.055)	(0.055)	(0.060)
Observations	107,312	100,689	70,592	85,037	57,363
R-squared	0.665	0.665	0.710	0.699	0.733
Controls	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Worker FE	Yes	Yes	Yes	Yes	Yes
Autonomy Score	No	Yes	No	No	Yes
Age Restriction	No	No	Yes	No	Yes
Education Restriction	No	No	No	Yes	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET conscience score is scaled from zero to one using min-max standardization. Controls are from Table 2. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, race/ethnicity, sex, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). Note that indicators for race/ethnicity and sex are omitted when controlling for worker fixed effects.

Second, the NLSY97 sample consists of workers who are relatively young to mid-career – 37 to 41 years of age in the most recently observed year. Therefore, many of the worker-occupation-year observations employed in the analysis are for workers who are relatively young and possibly still enrolled in school. In an attempt to identify worker-occupation-year observations that reflect more permanent employment endeavors, we limit our sample to workers aged 24 and over. As a result, the total number of observations drops to 70,592, but the estimated compensating differential reported in column (3) remains consistent with the baseline model in column (1). In other words, transitioning from an occupation where workers are most often asked to compromise their conscience to one where they are least

often asked is associated with approximately a 9% decrease in hourly compensation.

Third, we limit our observations to workers who have completed their highest educational degree (see Table 2). By focusing on workers who have finished their education, we aim to capture a more homogeneous group in terms of labor market experience. As noted in column (4), the estimated negative relationship between the conscience score and wages remains consistent with our baseline model in column (1), although the estimated coefficient decreased modestly from -0.091 in column (1) to -0.077 in column (4).

Finally, the last column combines all previous tests: adding the autonomy score, restricting the sample to workers aged 24 and over, and limiting the sample to those who have completed their highest educational degree. The resulting coefficient is -0.084, remaining statistically significant at the 1% level, with the sample size reduced to only 57,363. The negative relationship between the conscience score and wages remains robust and statistically significant across all sensitivity checks. Whether controlling for autonomy, restricting the sample to older or fully educated workers, a decrease in wages persists for workers in occupations where they are least often pressured to compromise their conscience in a job.[12]

3.3 Heterogeneous Effects by Age and Education

In this section, we investigate whether the price of conscience varies across age and educational attainment. Previous studies on compensating differentials for job-related risks or health hazards consistently show that older workers place a higher premium on risk compared to younger workers. Likewise, more educated workers tend to assign a greater premium to risk than their less educated counterparts (i.e., Arnould and Nichols, 1983). Table 5 presents the interaction effects of the conscience score with age and educational levels.

The results of the interaction effects of the conscience score with age are presented in column (1) of Table 5. First, it is evident that wages increase with age. Second, a higher conscience score is associated with lower compensation, reflected in the negative coefficient on the conscience score. Third, the estimated interaction effect shows that the negative association between the conscience score and wages becomes more pronounced as workers age (i.e., main effect + interaction effect). For illustrative purposes, consider two examples of workers aged 30 and 40, as shown in Figure 3, where we plot the predicted hourly compensation by age

[12] As previously discussed, occupations in the Natural Resource, Construction, and Maintenance sector are types of jobs where workers are least often pressured to compromise their conscience (see Figure 1). Most of these occupations are clustered at a raw conscience score of 4 and above, suggesting that they may drive our baseline results. Therefore, in Table A3 in the appendix, we exclude these occupations and re-estimate Equation 2, finding that the negative relationship between the conscience score and wages remains robust and statistically significant.

Table 5. Heterogeneity Estimates of O*NET Conscience Score vis-a-vis Log Real Hourly Compensation

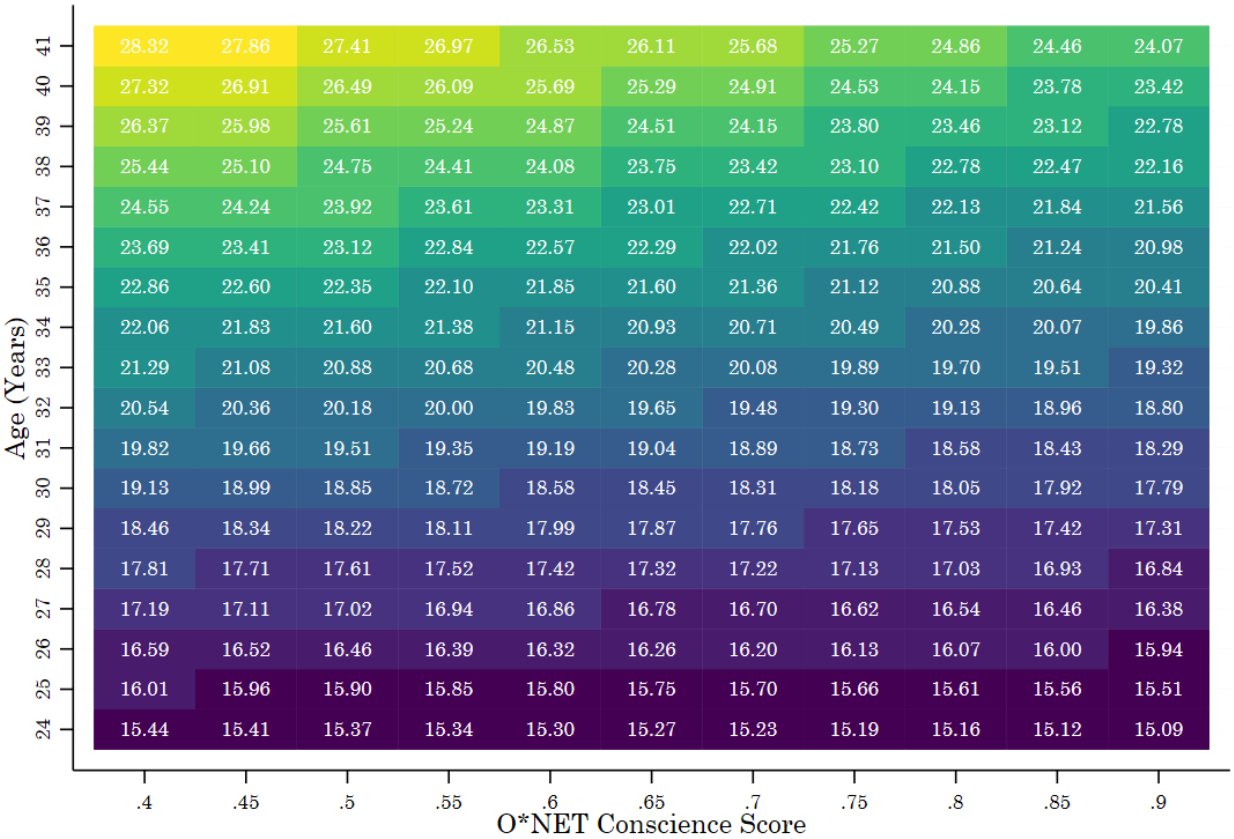
	(1)	(2)
Conscience Score (Z_{ij})	-0.098***	-0.046
	(0.010)	(0.037)
Conscience Score × Age	-0.016***	
	(0.001)	
Age	0.032***	
	(0.001)	
Conscience Score × Associates		-0.128**
		(0.062)
Conscience Score × Bachelors		-0.143***
		(0.050)
Conscience Score × Graduates		-0.291***
		(0.098)
Associates		0.159***
		(0.050)
Bachelors		0.268***
		(0.032)
Graduates		0.560***
		(0.053)
Constant	2.622***	2.218***
	(0.021)	(0.054)
Observations	107,312	107,312
R-squared	0.664	0.665
Controls	Yes	Yes
Year FE	Yes	Yes
Worker FE	Yes	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET conscience score is scaled from zero to one using min-max standardization. Age and conscience score are centered variables in column (2). Controls are from Table 2. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, race/ethnicity, sex, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). Note that indicators for race/ethnicity and sex are omitted when controlling for worker fixed effects.

and the conscience score, with all other model covariates set to their means. For 30-year-old workers, the predicted hourly compensation at the minimum conscience score is $19.13, while at the maximum conscience score, it is $17.29, resulting in a decrease in wages of about 9.6% per hour. For 40-year-old workers, the predicted hourly compensation at the minimum conscience score is $27.32, while at the maximum conscience score, it is $23.42, resulting in a compensating differential of about 22.4% per hour. Figure 3 shows that the willingness of workers to assume occupations with high conscience compromise *appears*[13] to decrease with

[13]However, the interaction effect between conscience score and age may not be solely due to workers getting

age. In other words, absent a market price for conscience, workers are evidently less willing to take on conscience-compromising work.



Notes: These predictions are from the estimated coefficient for the interaction between the conscience score and age in column (1) of Table 5. Both age and conscience score are centered variables. Hourly compensation is measured in 2020 U.S. dollars.

Figure 3. Predicted Hourly Compensation by Age and O*NET Conscience Score

Turning to column (2) of Table 5, we find consistent evidence with Arnould and Nichols (1983) and others that more educated workers tend to assign a greater premium to workplace disamenities than their less educated counterparts. For example, compared to workers with a high school diploma or less (the main effect), workers with an associate, bachelors, or graduate degree (interaction effects) are associated with a 16%, 17.2%, or 29% (main effect + interaction effects) decrease in hourly compensation, respectively. One possible explanation is that educated workers have better outside options or higher bargaining power, enabling them to negotiate higher compensation to accept conscience-compromising job.

3.4 Nonlinear Effects of Conscience Scores on Wages

Thus far, our empirical models have assumed a linear relationship between the conscience score and wages. However, it may be the case that the effect of occupational conscience

older. Other factors, such as potential changes in federal policies that correlate with conscience scores, could also influence this relationship. Therefore, readers should interpret our results with caution.

is nonlinear as observed with other (dis)amenities. Take for example the job attribute of exposure to life threatening hazards. Such disamenities require extra compensation but the absence of such occupational risks do not necessarily result in the loss of compensation (e.g., Leeth and Ruser, 2003; Guardado and Ziebarth, 2019). The same nonlinearity is present in the so-called *care penalty* literature, where workers in other-regarding occupations – those involving the provision of care to others, such as nursing, teaching, social work, or child care – tend to earn lower wages compared to workers in occupations requiring similar skills but not involved in the direct care for others. Workers in care occupations are said to derive intrinsic satisfaction from the nature of their work, with this non-monetary benefit pressing their wages lower. The care penalty operates asymmetrically, meaning that the absence of care work does not necessarily result in a wage premium (England et al., 2002; Hirsch and Manzella, 2015; Budig et al., 2019).

The compromise of conscience may function similarly – that is, the presence of compromise may require a premium but the relative absence of compromise may not induce penalty. To explore this potential nonlinearity, we consider a piecewise linear worker fixed-effects model that allows for different slopes across various ranges of the conscience score. This approach can help determine whether the wage effect changes beyond a certain threshold of moral compromise. For simplicity, we use the median conscience score as the threshold value (i.e., $\text{median}(Z_{ij}) = 0.65$) and construct two new variables, $\underline{Z_{ij}}$ and $\overline{Z_{ij}}$, based on the following:

$$\underline{Z_{ij}} = \begin{cases} Z_{ij}, & \text{if } Z_{ij} \leq \text{median}(Z_{ij}) \\ \text{median}(Z_{ij}), & \text{if } Z_{ij} > \text{median}(Z_{ij}) \end{cases}$$

$$\overline{Z_{ij}} = \begin{cases} 0, & \text{if } Z_{ij} \leq \text{median}(Z_{ij}) \\ Z_{ij} - \text{median}(Z_{ij}), & \text{if } Z_{ij} > \text{median}(Z_{ij}) \end{cases}$$

Using these newly defined variables, we re-estimate our worker fixed-effects model in Equation (2) by replacing Z_{ij} while all other terms remain unchanged from Equation (2). The updated model is as follows:

$$ln(w_{ijt}) = X'_{ijt}\beta + \alpha_1 \underline{Z_{ij}} + \alpha_2 \overline{Z_{ij}} + A'_{ij}\gamma + \eta_i + \delta_t + \epsilon_{ijt}. \tag{3}$$

Column (1) in Table 6 reports the estimated coefficients α_1 and α_2 from Equation (3). The results provide clear evidence of a nonlinear relationship between the conscience score and hourly compensation. Specifically, the significant negative effect in the lower range (i.e., $\underline{Z_{ij}}$) and the negligible effect in the higher range (i.e., $\overline{Z_{ij}}$) demonstrate that the relationship is not uniform across the conscience score vis-a-vis compensation.

To formally test for nonlinearity, we regress the log of real hourly compensation on the conscience score and the conscience score squared – a method common in the compensating wage differentials literature – while keeping all other terms in Equation (2) unchanged. The regression model is specified as follows:

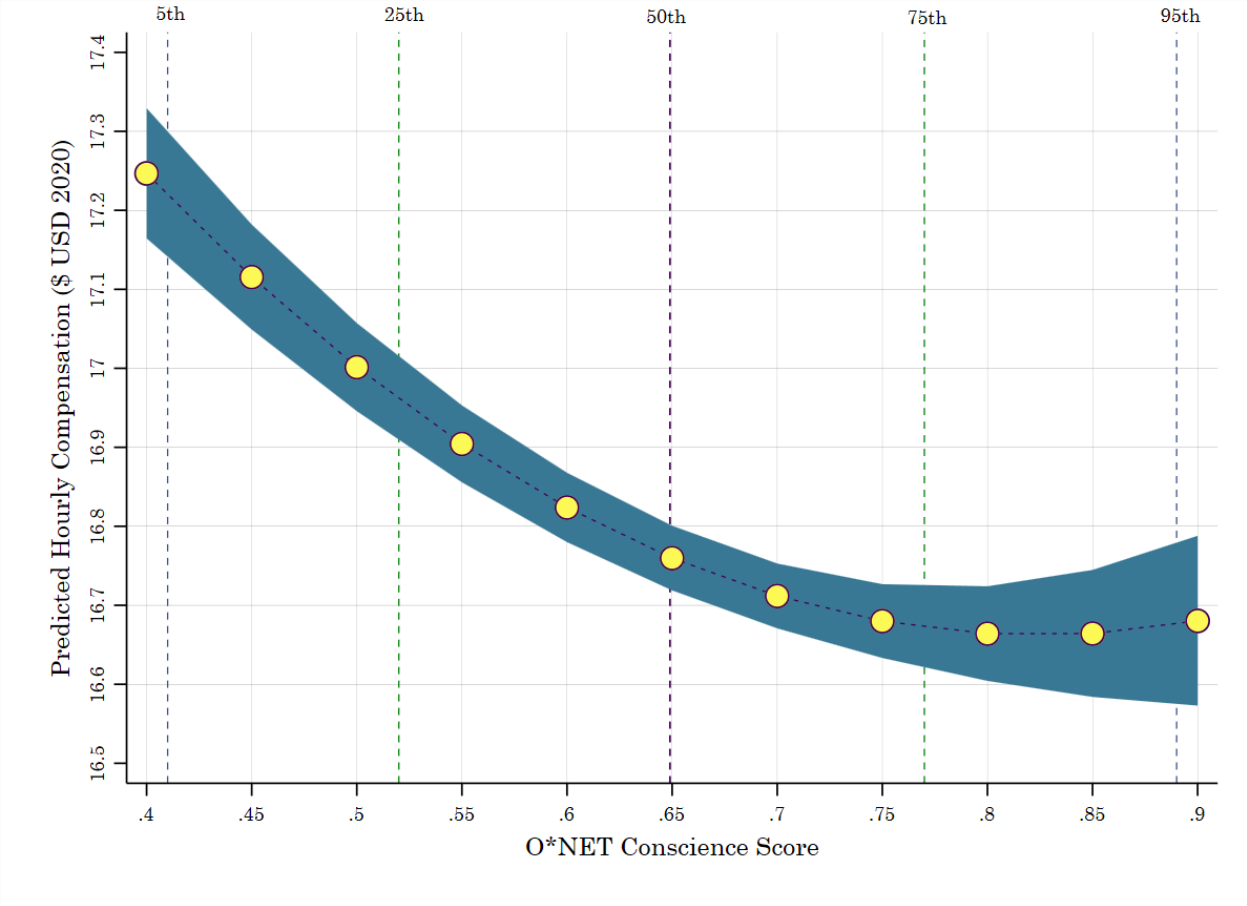
$$ln(w_{ijt}) = X'_{ijt}\beta + \alpha_1 Z_{ij} + \alpha_2 Z_{ij}^2 + A'_{ij}\gamma + \eta_i + \delta_t + \epsilon_{ijt}. \tag{4}$$

We then plot the resulting estimated relationship in Figure 4. First, column (2) of Table 6 further underscores this nonlinearity, as the coefficient on the conscience score squared is both positive and significant. Second, Figure 4 illustrates that for conscience scores below the median threshold, the slope is relatively steeper compared to conscience scores above the median threshold vis-a-vis wages. Moreover, the slope gradually flattens around the 75th percentile. These observations are consistent with our results in column (1) of Table 6, suggesting that the premium for the relative presence of compromise is greater than the penalty for the relative absence of compromise.

Table 6. Nonlinear Estimates of O*NET Conscience Score vis-a-vis Log Real Hourly Compensation

	(1)	(2)
Conscience Score Low ($\underline{Z_{ij}}$)	-0.137**	
	(0.055)	
Conscience Score High ($\overline{Z_{ij}}$)	-0.021	
	(0.062)	
Conscience Score (Z_{ij})		-0.315***
		(0.118)
Conscience Score × Conscience Score (Z_{ij}^2)		0.191**
		(0.094)
Constant	2.265***	2.303***
	(0.055)	(0.058)
Observations	107,312	107,312
R-squared	0.665	0.665
Controls	Yes	Yes
Year FE	Yes	Yes
Worker FE	Yes	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET conscience score is scaled from zero to one using min-max standardization. Controls are from Table 2. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, race/ethnicity, sex, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). Note that indicators for race/ethnicity and sex are omitted when controlling for worker fixed effects.



Notes: The curve is the predicted relationship from Equation 4, with all other model covariates set to their means. Shaded area denotes a 95 percent confidence interval.

Figure 4. Predicted Hourly Compensation ($ USD 2020)

4 Discussion and Conclusion

Nora Szech's experimental and theoretical research on *morals and markets* challenged the philosophical view that markets are morally indifferent. Her work highlights how markets shape and possibly degrade the moral convictions of market participants. Extending her experimental insights to real-world labor markets, we empirically examined how markets differentially compensate jobs that require workers to compromise their moral *sense of right and wrong*, allowing us to quantify the "price of conscience" in labor market transactions.

Our analysis suggests that there is, indeed, a premium associated with the burdening of conscience. Under the most stringent restrictions – in column (5), Table 4 – we find workers gain about 9% in hourly compensation in going from the least to the most compromising occupations. This premium appears to increase with age and education, suggesting that experienced and educated workers require a higher price for compromising their conscience. Similar to other job (dis)amenities, we find evidence of a nonlinear relationship between

wages and conscience, where the premium for the compromise of conscience is greater than the penalty (or non-wage benefit) of assuming work that aligns with one's moral convictions.

Before closing, there are important limitations to our study worth reiterating. Because our sample consists only of workers who are relatively young to mid-career, our estimates may not fully represent the broader economy or workers at later career stages. To address this concern, we examined how wages evolve over time within our sample by including an interaction between the conscience score and age in our analysis. This allows us to assess whether the relationship between conscience scores and wages changes as workers progress in their careers, providing some confidence that our results capture the progression of wage effects related to conscience scores over time.

As previously detailed, the use of worker fixed-effects may not fully account for time-varying residual confounders that influence both the selection into conscience-compromising jobs and wage outcomes. While we attempt to mitigate this issue by controlling for observed time-varying worker characteristics, unobserved time-varying factors could still bias our estimates. This limitation restricts our ability to establish a definitive causal relationship between conscience scores and wages. Thus, this study underscores the need for future research to disentangle the causal pathways, particularly concerning unobserved confounding variables in real-world labor market context. Despite these caveats, our findings provide novel evidence of how markets appear to systematically reward workers for doing things they define as wrong.

Leveraging coefficients from column (1) in Table 5, and assuming that a worker transitions permanently from an occupation in 75th (0.52) to 25th percentile (0.77) on the conscience score, one can extrapolate the price of conscience over an archetypal career course from age 24 to 64. Holding all else constant and assuming full employment at 40 hours per week and 52 weeks per year, we sum differences (between the 25th and 75th percentile in conscience score) in the predicted hourly compensation at each age over the hypothetical career span to get an estimate of $292,476 (95% CI: $159,914 to $425,294) in 2020 U.S. dollars.[14] One may interpret our quantitative estimates of the price of conscience as conservative. Just as workers place different prices on the risk of mortality, workers place different prices on their conscience. The marginal worker hired "to do things that go against their sense of right and wrong" places a smaller price on it than an inframarginal worker who likely values their moral integrity more highly.

[14]The lower bound estimate of $159,914 is calculated as the sum of the upper bound of predicted compensation at the 75th percentile of our conscience score for each age minus the lower bound of predicted compensation at the 25th percentile of the conscience score for each age. The upper bound estimate of $425,294 involves the opposite logic.

Compensating differentials for other job disamenities – like assuming the odious work of the butcher in Adam Smith's foundational statement on the wages of labor – increase total surplus. However, compensating differentials for moral disamenities arguably do not. It is efficient to incentivize and sort workers into jobs that are physically demanding, or schedule inflexible, or associated with higher mortality risk (Lavetti, 2023) because these disamenities are borne almost entirely by workers themselves. However, it is not similarly efficient to sort workers into jobs who require the least possible extra compensation *to do things that go against their sense of right and wrong* if those things impose measurable and non-trivial costs on others. Matching labor markets in human conscience might therefore function to increase the quantity of harm imposed on others.

A conclusion from Szech's experimental studies is that markets may erode the moral convictions of market participants by inoculating one's sense of personal responsibility (Falk and Szech, 2013) and inviting "consequentialist" rationalizations for actions that are manifestly wrong (Falk et al., 2020). The capitulation of "deontological moral reasoning" – an approach to ethics that focuses on the inherent rightness or wrongness of one's actions, regardless of their consequences – happens in market contexts, Falk et al. (2020) argue, because market participants see their actions as non-pivotal and ultimately replaceable.

While the price of conscience appears to increase with age and level of educational attainment, one mustn't ignore the inverse conclusion that younger, less experienced workers and especially those with a high school diploma or less, are differentially compensated least for the compromise of conscience. Insofar as one accepts the philosophical claims that *existence precedes essence* or that *consciousness (our thoughts, values and beliefs) follows from being*, markets may, in effect, most corrode the moral sensibilities of the most vulnerable among us – a kind of market stratification in the trade of conscience.

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Appendix

Table A1. NLSY97 Occupations (2002 Census Codes) That Were Manually Matched to O*NET Occupations (2018 SOC Codes)

NLSY97 Occupation	Code	O*NET SOC Occupation	Code
Public Relations Managers	60	Advertising and Promotions Managers	11-2011.00
		Administrative Services Managers	11-3012.00
Financial and Investment Analysts	840	Personal Financial Advisors	13-2052.00
Network Systems and Data Communications Analysts	1110	Computer Network Support Specialist	15-1231.00
Entertainers and Performers, Sports and Related Workers, All Other	2760	Musicians and Singers	27-2042.00
		Dancers	27-2031.00
		Athletes and Sports Competitors	27-2021.00
Media and Communication Equipment Workers, All Other	2960	Audio and Video Technicians	27-4011.00
		Sound Engineering Technicians	27-4014.00
		Camera Operators, Television, Video, and Film	27-4031.00
Emergency Medical Technicians and Paramedics	3400	Firefighters	33-2011.00
Medical Records and Health Information Technicians	3510	Medical Transcriptionists	31-9094.00
First-Line Supervisors of Protective Service Workers, All Other	3730	First-Line Supervisors of Correctional Officers	33-1011.00
		First-Line Supervisors of Police and Detectives	33-1012.00
		First-Line Supervisors of Firefighting and Prevention Workers	33-1021.00
Counter Attendant, Cafeteria, Food Concession, and Coffee Shop	4060	Fast Food and Counter Workers	35-3023.00
		Baristas	35-3023.01
Personal Care Aides	4610	Childcare Workers	39-9011.00
		Nannies	39-9011.01
Personal Care and Service Workers, All Other	4650	Childcare Workers	39-9011.00
		Nannies	39-9011.01
		Exercise Trainers and Group Fitness Instructors	39-9031.00
Sales and Related Workers, All Other	4960	Telemarketers, Door-to-Door Sales Workers, News and Street Vendors, and Related Workers	41-9041.00
Communications Equipment Operators, All Other	5030	Switchboard Operators, Answering Service	43-2011.00
		Telephone Operators	43-2021.00
Information and Record Clerks, All Other	5420	Receptionists and Information Clerks	43-4171.00
Stock Clerks and Order Fillers	5620	Shipping, Receiving, and Inventory Clerks	43-5071.00
Computer Operators	5800	Data Entry Keyers	43-9021.00
		Word Processors and Typists	43-9022.00
		Office Clerks, General	43-9061.00
		Office Machine Operators, Except Computer	43-9071.00
Office and Administrative Support Workers, All Other	5930	Office Clerks, General	43-9061.00
Computer Control Programmers and Operators	7900	Extruding and Drawing Machine Setters, Operators, and Tenders, Metal and Plastic	51-4021.00
		Forging Machine Setters, Operators, and Tenders, Metal and Plastic	51-4022.00
		Setters, Operators, and Tenders, Metal and Plastic	51-4031.00
		Drilling and Boring Machine Tool Setters, Operators, and Tenders, Metal and Plastic	51-4032.00
Metal Workers and Plastic Workers, All Other	8220	Layout Workers, Metal and Plastic	51-4192.00
		Plating Machine Setters, Operators, and Tenders, Metal and Plastic	51-4193.00
Bookbinders and Bindery Workers	8230	Print Binding and Finishing Workers	51-5113.00
Job Printers	8240	Printing Press Operators	51-5112.00
Textile, Apparel, and Furnishings Workers, All Other	8460	Fabric and Apparel Patternmakers	51-6092.00
		Upholsterers	51-6093.00

Continued on next page

NLSY97 Occupation	Code	O*NET SOC Occupation	Code
Woodworkers, All Other	8550	Cabinetmakers and Bench Carpenters	51-7011.00
		Furniture Finishers	51-7021.00
Miscellaneous Plant and System Operators	8630	Water and Wastewater Treatment, Plant and System Operators	51-8031.00
		Chemical Plant and System Operators	51-8091.00
Transportation Attendants	4550	Flight Attendants	53-2031.00
		Passenger Attendants	53-6061.00
Taxi Drivers and Chauffeurs	9140	Bus Drivers, Transit and Intercity	53-3052.00
Motor Vehicle Operators, All Other	9150	Heavy and Tractor-Trailer Truck Drivers	53-3032.00
		Light Truck Drivers	53-3033.00
Other transportation workers	9420	Traffic Technicians	53-6041.00
Material Moving Workers, All Other	9750	Laborers and Freight, Stock, and Material Movers, Hand	53-7062.00
		Tank Car, Truck, and Ship Loaders	53-7121.00

Table A2. Full Results for Column (4) of Table 3

	Log Real Hourly Compensation
Conscience Score (Z_{ij})	-0.091***
	(0.034)
Degree (Reference = $\leq$ High School)	
Associates	0.080***
	(0.017)
Bachelors	0.187***
	(0.011)
Graduates	0.410***
	(0.017)
Cum. Years NILF	-0.017***
	(0.001)
Cum. Years in Unemployment	-0.057***
	(0.004)
Union Status	0.102***
	(0.005)
Self-Employed	0.148***
	(0.023)
Work Hours per Week	0.001
	(0.001)
Job Tenure (Years)	0.010***
	(0.001)
Other O*NET Occupational Attributes	
Hazard Index	0.056
	(0.035)
Skills Index	0.255***
	(0.036)
Care Index	-0.098
	(0.068)
Social Index	0.027
	(0.086)
Constant	2.249***
	(0.052)
Observations	107,312
R-squared	0.665
Year FE	Yes
Worker FE	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET indices and scores are scaled from zero to one using min-max standardization. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). NILF implies not-in-the-labor-force.

Table A3. Sensitivity Estimates of O*NET Conscience Score Without Occupations in Natural Resource, Construction, and Maintenance Sector (6000-7620)

	Log Real Hourly Compensation
Conscience Score (Z_{ij})	-0.102***
	(0.042)
Degree (Reference = $\leq$ High School)	
Associates	0.082***
	(0.019)
Bachelors	0.186***
	(0.012)
Graduates	0.417***
	(0.018)
Cum. Years NILF	-0.016***
	(0.001)
Cum. Years in Unemployment	-0.057***
	(0.005)
Union Status	0.092***
	(0.006)
Self-Employed	0.148***
	(0.023)
Work Hours per Week	0.001
	(0.001)
Job Tenure (Years)	0.010***
	(0.001)
Other O*NET Occupational Attributes	
Hazard Index	0.054
	(0.043)
Skills Index	0.270***
	(0.049)
Care Index	-0.121
	(0.079)
Social Index	0.036
	(0.099)
Constant	2.299***
	(0.064)
Observations	88,734
R-squared	0.682
Year FE	Yes
Worker FE	Yes

Notes: Standard errors clustered by occupation level denoted in parentheses. Significance is represented as follows: *** p<0.01, ** p<0.05, * p<0.1. O*NET indices and scores are scaled from zero to one using min-max standardization. Additional controls include indicators for urbanicity, MSA, Census region, industry-specific fixed-effects, and age (e.g., 18-21, 22-25, 26-29, 30-33, 34-37, and 38-41). NILF implies not-in-the-labor-force.